Physics-Informed Neural Ensemble Data Assimilation via Conditional Flow Matching

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Key Points:

- Physics-Informed Generative Modeling: Bridges variational data assimilation and conditional flow matching to enable physics-informed sampling of the posterior for complex space-time dynamics.
- Tweedie-Inspired Architecture: Derives a theoretically grounded neural operator based on Tweedie's Formula, allowing for the explicit decomposition of the conditional expectation into Gaussian and non-Gaussian components.
- Demonstrated Scalability: Validates the approach across multiple scales, ranging from toy Lorenz-96 systems to intermediate-complexity quasi-geostrophic (QG) flows and global sea surface dynamics.

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Abstract

Data assimilation (DA) is central to geosciences for the reconstruction of space-time state variables from partial and noisy observations. While current state-of-the-art DA systems rely on large-scale variational or ensemble-based schemes, they often face computational bottlenecks and struggle to characterize the full non-Gaussian nature of the posterior distribution in high-dimensional systems. In this work, we introduce 4DVarNet-FM, an end-to-end neural ensemble DA framework that bridges variational data assimilation and conditional flow matching (CFM). By parameterizing the sampling process through trainable stochastic interpolants, our approach explicitly incorporates physical priors within the neural architecture. We demonstrate that the conditional expectation operator of the posterior can be theoretically derived using a Tweedie-inspired formulation, enabling the unrolled neural solver to distinguish between linear Gaussian updates—analogous to traditional Ensemble Kalman Filter schemes—and non-linear structural residuals. Through experiments on Lorenz-96 dynamics, 2D quasi-geostrophic flows, and global-scale sea surface dynamics, we show that 4DVarNet-FM provides a computationally efficient and physically consistent solution for posterior sampling. Our findings support the relevance of integrating variational formulations into generative neural architectures to address the complexity of real-world Earth system dynamics.

Plain Language Summary

1 Introduction

Data assimilation is a cornerstone of modern geosciences, providing the essential framework for estimating the state of high-dimensional, chaotic dynamical systems from sparse, partial, and noisy observations. Despite their critical role in applications ranging from numerical weather prediction to oceanography, state-of-the-art DA systems face significant challenges. Conventional variational and ensemble-based schemes often struggle with the high computational cost of large-scale implementations, the difficulty of characterizing non-Gaussian posterior distributions, and the necessity of prescribing explicit, often idealized, physical priors. ML for Data assimilation.

In response to these limitations, recent years have seen a surge in machine learning (ML) approaches for DA. Early efforts focused on leveraging ML to improve specific components of existing DA pipelines, such as through state-space reconstruction or the enhancement of representational models. More recently, the field has transitioned toward the end-to-end design and calibration of full DA pipelines, such as 4DVarNet, which demonstrate the potential for neural architectures to learn effective mapping operators directly from data. While these neural approaches have shown promise on toy models and intermediate-complexity systems, their ability to scale to real-world Earth system dynamics—where physical consistency and uncertainty quantification are paramount—remains a subject of intensive research.

In this work, we explore an end-to-end neural ensemble DA scheme that bridges the rigorous formulation of variational data assimilation and state-of-the-art generative modeling, specifically leveraging conditional flow matching (FM) approaches. Our contributions are three-fold:

- We introduce a class of end-to-end neural DA architectures that utilize state-of-the-art U-Net-based flow matching. These architectures encompass, as specific parameterizations, a variety of inversion schemes, including 4DVar, score-based methods, and previous 4DVarNet formulations.
- We demonstrate, through diverse case studies—including Lorenz-96 dynamics, 2D+t quasi-geostrophic (QG) flows, and global-scale sea surface dynamics—how the pro-

posed approach successfully scales to address the computational complexity of Earth system dynamics.
- Our experiments support the relevance of an underlying variational data assimilation formulation, enabling the neural solver to explicitly benefit from prior knowledge regarding the considered observing system and physical dynamics.

2 Background and related work

This section provides the mathematical and conceptual foundations necessary to bridge variational data assimilation and generative modeling. We first review the principles of flow matching and stochastic interpolants, before positioning our proposed framework within the evolving landscape of neural data assimilation.

2.1 Flow matching

For random fields $\mathbf{x}_0 \sim \pi_0$ and $\mathbf{x}_1 \sim \pi_1$, we define a time-varying field $\mathbf{x}_\tau$ as:

$$\forall \tau \in [0,1], \mathbf{x}_\tau = \alpha_\tau \cdot \mathbf{x}_0 + \beta_\tau \cdot \mathbf{x}_1 \tag{1}$$

where π_1 is the target distribution we aim to sample and π_0 is the chosen distribution, usually a centered normal distribution with a unit spherical covariance (Lipman et al., 2023; Albergo et al., 2025). Following (Lipman et al., 2023; Albergo et al., 2025), one can derive ODE and SDE to sample π_1 from π_0 of the form:

$$\begin{cases} \mathbf{x}_0 \sim \mathcal{N}(0, \mathbb{1}) \\ \\ \dfrac{d\mathbf{x}_\tau}{d\tau} = \dot{\alpha}_\tau \mathbf{x}_\tau + (\dot{\beta}_\tau - \beta_\tau \dfrac{\dot{\alpha}_\tau}{\alpha_\tau}) \mathbb{E}[\mathbf{x}_1 | \mathbf{x}_\tau] \end{cases} \tag{2}$$

In recent years, Flow Matching (FM) has emerged as a state-of-the-art framework for generative modeling, demonstrating exceptional performance across diverse fields including signal processing, image generation, and complex biological and physical systems (Lipman et al., 2024; Holderrieth & Erives, 2026). By learning continuous-time transformations—typically parameterized as neural ODEs—between a simple reference distribution and a target data distribution, FM provides a computationally-efficient alternative to traditional diffusion-based approaches, as it eliminates the need for simulation during the training phase (Lipman et al., 2024; Ahamed & Haber, 2026).

In the geosciences, FM is increasingly recognized for its ability to model complex, high-dimensional Earth system dynamics. Recent advancements demonstrate its effectiveness in tasks such as multi-scale climate emulation and the reconstruction of turbulent flow fields, where FM enables the generation of physically consistent, high-fidelity data while simultaneously quantifying predictive uncertainty (Parikh et al., 2025; ?, ?). Furthermore, the framework has proven robust in addressing ill-posed inverse problems in imaging and fluid dynamics, providing a principled method for synthesizing multiple plausible realizations that reflect both informative and stochastic components of the underlying system (Ahamed & Haber, 2026; Parikh et al., 2025).

2.2 Variational Data assimilation and Uncertainty Quantification

Data assimilation addresses the reconstruction of the state variable $\mathbf{x}$ given some available observation $\mathbf{y}$. From a theoretical point of view, we can state the design of a variational data assimilation system as a bi-level optimization problem:

$$\min_\theta \mathbb{E}_{\mathbf{x}^*, \mathbf{y}} [\|\mathbf{x}^* - \hat{\mathbf{x}}\|^2] \quad \text{s.t.} \quad \hat{\mathbf{x}} = \arg\min_{\mathbf{x}} \mathcal{J}_\theta(\mathbf{x}, \mathbf{y}) \tag{3}$$

where $\mathbf{x}^*$ is the true state, $\hat{\mathbf{x}}$ the reconstructed state and $\mathcal{J}_\theta$ the variational cost with parameters θ. As detailed in Section 3, the variational cost usually splits as the sum of several terms:

$$\mathcal{J}_\theta(\mathbf{x},\mathbf{y}) = \mathcal{J}_{obs}(\mathbf{x},\mathbf{y}) + \mathcal{J}_{reg}(\mathbf{x}) + \mathcal{J}_{bg}(\mathbf{x}) \tag{4}$$

with $\mathcal{J}_{obs}(\mathbf{x},\mathbf{y})$ the observation term, $\mathcal{J}_{reg}(\mathbf{x})$ the regularization term stating a prior on the considered dynamics and $\mathcal{J}_{bg}(\mathbf{x})$ a background term stating a first guess on the solution.

When dealing with linear-quadratic variational cost associated with covariance-based priors, the solution of the inner minimization is the best linear unbiased estimator, also referred to as optimal interpolation (Carrassi et al., 2018). Regarding non-Gaussian and non-linear systems, a large literature has dealt with the inner minimization of the variational cost (Carrassi et al., 2018). However, solving the bi-level optimization is in general intractable. This implies that one cannot guarantee in general the optimality of the data assimilation system.

Hereafter, following (Carrassi et al., 2018; Fablet et al., 2021) we will consider Gaussian observation term $\mathcal{J}_{obs}(\mathbf{x},\mathbf{y}) = \|\mathbf{y}-\mathcal{H}(\mathbf{x})\|_R^2$ with $\mathcal{H}$ the observation operator and R the observation covariance, and a Gaussian background term $\mathcal{J}_{bg}(\mathbf{x}) = \|\mathbf{x} - \mathbf{x}_B\|_B^2$ with $\mathbf{x}_B$ the background mean and B the background covariance. Regarding regularization term, we consider a generic form $\mathcal{J}_{reg}(\mathbf{x}) = \|\mathbf{x}-\Phi(\mathbf{x})\|_Q^2$ which accounts for both model-based and data-driven regularisation terms (Fablet et al., 2021).

In recent years, the intersection of machine learning (ML) and data assimilation has evolved toward the development of 'Neural DA'—a paradigm where neural architectures are trained to act as end-to-end solvers for the variational assimilation problem. Departing from the traditional approach of using ML solely as a surrogate for specific components, this trend focuses on unrolling the iterative minimization of $\mathcal{J}_\theta(\mathbf{x},\mathbf{y})$ within the neural network's forward pass, as pioneered by Fablet et al. (Fablet et al., 2021). Such architectures leverage the representational power of deep learning to effectively manage the high-dimensional, non-Gaussian structures inherent in Earth system dynamics. Concurrently, the field has seen an influx of score-based and generative methodologies, with Rozet and Louppe (Rozet & Louppe, 2023) demonstrating the potential of score-based DA to provide robust posterior sampling. These efforts are complemented by large-scale data-driven forecasting models (Lam et al., 2023; Aouni et al., n.d.; Pathak et al., 2022), which emphasize the shift toward purely data-driven high-resolution modeling. By parameterizing the solver Ψ_θ as a deep learning operator, these frameworks sidestep the intractability of classical bi-level optimization, enabling the autonomous discovery of physical priors directly from data. As we explore in this work, 4DVarNet-FM further bridges this gap, unifying the rigor of variational principles with the generative capabilities of conditional flow matching.

3 Proposed approach

3.1 Considered Bayesian formulation

We address data assimilation problems for the reconstruction of a space-time state $\mathbf{x}_1$ given observations $\mathbf{y}$ for a given space-domain $\mathcal{T} \times \Omega$. Within a Bayesian formulation, we introduce the following state-space formulation for a predefined discretization of the space-time domain:

$$\left\{\begin{array}{lcl} \mathbf{x}_0 & \sim & \mathcal{N}(0, \mathbb{1}) \\ \mathbf{x}_1 & \sim & \pi_1 \text{ with } \pi_1(\mathbf{x}_1) \propto \exp\left[-\frac{1}{2}\|\mathbf{x}_1 - \Phi(\mathbf{x}_1)\|_{\mathbf{Q}}^2\right] \\ \mathbf{y} & = & \mathcal{H}(\mathbf{x}_1) + \epsilon_{obs} \text{ with } \epsilon_{obs} \sim \mathcal{N}(0, \mathbf{R}) \\ \mathbf{x}_\tau & = & \alpha_\tau \cdot \mathbf{x}_0 + \beta_\tau \cdot \mathbf{x}_1, \quad \forall \tau \in [0,1] \end{array}\right. \tag{5}$$

where operator Φ and covariance $\mathbf{R}$ define the energy-based prior on state $\mathbf{x}_1$. $\mathcal{H}$ is the observation operator and $\mathcal{N}(0, \mathbf{R})$ a normal distribution with covariance $\mathbf{R}$. α_τ and β_τ are scalar time-varying functions such that $\alpha_0 = \alpha_1 = 1$ and $\alpha_1 = \beta_0 = 0$. Interestingly, we can derive the joint likelihood $p(\mathbf{x}_1, \mathbf{x}_\tau, \mathbf{y})$ as

$$
\begin{aligned}
p(\mathbf{x}_1, \mathbf{x}_\tau, \mathbf{y}) &\propto \exp\left[-\frac{1}{2}\mathcal{J}_\tau(\mathbf{x}_1, \mathbf{x}_\tau, \mathbf{y})\right] \\
\text{with } \mathcal{J}_\tau(\mathbf{x}_1, \mathbf{x}_\tau, \mathbf{y}) &= \|\mathbf{y} - \mathcal{H}(\mathbf{x}_1)\|^2_{\mathcal{R}} + \|\mathbf{x}_1 - \phi(\mathbf{x}_1)\|^2_{\mathbf{Q}} + \frac{\beta_\tau^2}{\alpha_\tau^2 \sigma_0^2}\left\|\beta_\tau^{-1}\mathbf{x}_\tau - \mathbf{x}_1\right\|^2,
\end{aligned}
\tag{6}
$$

This parameterization is similar to that of a variational data assimilation in (**??**) where $\beta_\tau^{-1}\mathbf{x}_\tau$ acts as the mean of a time-dependent background term with covariance $\mathbf{B} = \alpha_t^2\sigma_0^2\mathbb{1}$. Interestingly, we can also derive the score ∇_τ

From flow matching framework (**??**) applied to $\mathbf{x}_1|\mathbf{y}$, we also derive an ODE system to sample the posterior of state $\mathbf{x}_1$ conditionally to observation $\mathbf{y}$ from the proposal distribution of $\mathbf{x}_0$

$$
\left\{
\begin{array}{lcl}
\mathbf{x}_0 & \sim & \mathcal{N}(0, \mathbb{1}) \\
\dfrac{d\mathbf{x}_\tau}{d\tau} & = & \dfrac{\dot{\alpha}_\tau}{\alpha_\tau}\mathbf{x}_\tau + \left(\dot{\beta}_\tau - \beta_\tau \dfrac{\dot{\alpha}_\tau}{\alpha_\tau}\right)\mathbb{E}(\mathbf{x}_1|\mathbf{x}_\tau, \mathbf{y})
\end{array}
\right.
\tag{7}
$$

This ODE explicitly relies on the computation of the conditional expectation $\mathbb{E}(\mathbf{x}_1|\mathbf{x}_\tau, \mathbf{y})$. As the conditional expectation is the optimizee of a MSE criterion, we build on the analogy between (**??**) and (6), and draw a connection between solving bi-level data assimilation problem (**??**) using neural solvers as proposed in (Fablet et al., 2021) and conditional-flow-matching-based sampling schemes to introduce unrolled neural architecturs to approximate conditional expectation $\mathbb{E}[\mathbf{x}_1|\mathbf{x}_\tau, \mathbf{y}]$.

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3.2 Tweedie-Inspired decomposition of the conditional expectation operator $\mathbb{E}[\mathbf{x}_1|\mathbf{x}_\tau, \mathbf{y}]$

To theoretically ground our neural approach, we analyze the structure of the optimal conditional expectation operator. To generalize the conditional expectation $\mathbb{E}[\mathbf{x}_1|\mathbf{x}_\tau, \mathbf{y}]$ to arbitrary distributions, we leverage Tweedie's Formula (Rozet & Louppe, 2023; ?, ?). Recall the flow-matching latent path $\mathbf{x}_\tau = \beta_\tau\mathbf{x}_1 + \alpha_\tau\mathbf{x}_0$, which can be rewritten as:

$$
\frac{\mathbf{x}_\tau}{\beta_\tau} = \mathbf{x}_1 + \frac{\alpha_\tau}{\beta_\tau}\mathbf{x}_1 \tag{8}
$$

Tweedie's Formula yields the exact minimum mean square error (MMSE) estimator for $\mathbf{x}_1$:

$$
\mathbb{E}[\mathbf{x}_1|\mathbf{x}_\tau, \mathbf{y}] = \frac{\mathbf{x}_\tau}{\beta_\tau} + \frac{\alpha_\tau^2}{\beta_\tau}\nabla_{\mathbf{x}_\tau}\log p(\mathbf{x}_\tau|\mathbf{y}) \tag{9}
$$

Separating the true score function into a linear Gaussian-equivalent component and a non-linear structural residual $\mathcal{R}_{score}$:

$$
\nabla_{\mathbf{x}_\tau}\log p(\mathbf{x}_\tau|\mathbf{y}) = -\Sigma_\tau^{-1}(\mathbf{x}_\tau - \beta_\tau\boldsymbol{\mu}_{1|\mathbf{y}}) + \mathcal{R}_{score}(\mathbf{x}_\tau, \mathbf{y}, \tau) \tag{10}
$$

where $\boldsymbol{\mu}_{1|\mathbf{y}} = \mathbb{E}[\mathbf{x}_1|\mathbf{y}]$, and $\Sigma_\tau = \beta_\tau^2\Sigma_{1|\mathbf{y}} + \alpha_\tau^2\mathbb{I}$. The conditional expectation resolves into:

$$
\mathbb{E}[\mathbf{x}_1|\mathbf{x}_\tau, \mathbf{y}] = \underbrace{(\mathbb{1} - \beta_\tau K_\tau)\boldsymbol{\mu}_{1|\mathbf{y}} + K_\tau\mathbf{x}_\tau}_{\text{Optimal Gaussian Linear Update}} + \underbrace{\frac{\alpha_\tau^2}{\beta_\tau}\mathcal{R}_{score}(\mathbf{x}_\tau, \mathbf{y}, \tau)}_{\alpha_\tau\beta_\tau\Psi^{NG}(\mathbf{x}_\tau, \mathbf{y}, \tau)} \tag{11}
$$

with the gain matrix $K_\tau = \beta_\tau \left(\beta_\tau^2 \mathbb{I} + \alpha_\tau^2 \Sigma_{1|\mathbf{y}}^{-1}\right)^{-1}$. In the Gaussian limit, this gain K_τ recovers the structure of the Ensemble Kalman Filter (EnKF) update, where the ensemble covariance evolves to optimally weight the innovation. Unlike standard EnKF, however, this flow-matching operator adaptively adjusts the gain throughout the latent path $\tau \in [0, 1]$. By introducing the non-linear residual Ψ^{NG}, our framework effectively extends the linear subspace projection of EnKF to capture complex, non-Gaussian posterior structures. This formulation satisfies the expected boundary conditions:

- **Initialization ($\tau = 0$):** Since $\beta_0 = 0$ and $\alpha_0 = 1$, the latent state $\mathbf{x}_0$ is purely Gaussian. The score is exactly linear ($\mathcal{R}_{score} = 0$) and the gain matrix $K_0 = 0$. Consequently, the estimator collapses to the unconditioned posterior mean $\boldsymbol{\mu}_{1|\mathbf{y}}$.
- **Target Convergence ($\tau = 1$):** Since $\beta_1 = 1$ and $\alpha_1 = 0$, the prefactor $\alpha_\tau^2/\beta_\tau \to 0$, eliminating the non-linear residual Ψ_{NG}. Simultaneously, the gain matrix converges to the identity ($K_1 = \mathbb{I}$), forcing the estimator to perfectly retrieve the target state $\mathbf{x}_1$.

3.3 Neural solver for the conditional expectation operator $\mathbb{E}[\mathbf{x_1}|\mathbf{x_\tau}, \mathbf{y}]$

As discussed in Section 2, end-to-end neural Data Assimilation (DA) schemes (Fablet et al., 2021; Boudier et al., 2020; ?, ?) have emerged as a state-of-the-art paradigm for solving reconstruction problems in space-time dynamics. These schemes re-parameterize the bi-level optimization problem (**??**) by learning a neural operator Ψ_θ that approximates the conditional expectation $\mathbb{E}[\mathbf{x}_1|\mathbf{x}_\tau, \mathbf{y}]$ as follows:

$$\min_\theta \mathbb{E}_{\mathbf{x}_1, \mathbf{x}_\tau, y}[\|\mathbf{x}_1 - \hat{\mathbf{x}}_1\|^2] \quad \text{s.t. } \hat{\mathbf{x}}_1 = \Psi_\theta(\mathbf{y}, \mathbf{x}_\tau, \tau) \tag{12}$$

While standard flow matching typically employs a "black-box" one-step U-Net approximation to map conditioning variables to the target state (Lipman et al., 2023, 2024), we advocate for unrolled neural architectures that mimic the iterative refinement of a variational solver (Carrassi et al., 2018). By drawing a direct analogy between the variational cost $\mathcal{J}_\tau$ in (6) and 4DVar cost (**??**), we formulate Ψ_θ as an iterative operator that is physically constrained by the underlying energy-based posterior. This unrolled approach offers a distinct advantage over one-step mappings: it decomposes the complex task of approximating the state into a sequence of learned, physics-informed updates, allowing the model to refine its estimate across a controlled trajectory rather than relying on a single, high-dimensional non-linear mapping. As detailed below, we combine such unrolled neural architectures and Tweedie-inspired decomposition (11) for our neural parameterization of operator Ψ_θ.

Tweedie-inspired neural parameterization for neural operator Ψ_θ: From decomposition (11), we consider an additive parameterization of operator Ψ_θ in (12) according to a neural solver, denoted by $\bar{\Psi}_{\bar{\theta}}$, for the estimation of $\mathbb{E}[\mathbf{x}_1|\mathbf{y}]$, and a second neural solver, Ψ_θ^{NG} addressing the non-Gaussian term in (11). As an approximation of $\mathbb{E}[\mathbf{x}_1|\mathbf{y}]$, operator relies on the following training problem:

$$\min_{\bar{\theta}} \mathbb{E}_{\mathbf{x}_1, \mathbf{y}}[\|\mathbf{x}_1 - \hat{\mathbf{x}}_1\|^2] \quad \text{s.t. } \hat{\mathbf{x}}_1 = \bar{\Psi}_{\bar{\theta}}(\mathbf{y}) \tag{13}$$

The overall parameterization of operator Ψ_θ with $\theta = (\bar{\theta}, \tilde{\theta})$ is given by: $\forall \tau, \mathbf{x}_\tau, \mathbf{y}$,

$$\Psi_\theta(\mathbf{y}, \mathbf{x}_\tau, \tau) = \left(\mathbb{1} - \beta_\tau \tilde{K}_\tau\right) \bar{\Psi}_{\bar{\theta}} + \tilde{K}_\tau \mathbf{x}_\tau + \Psi_{\tilde{\theta}}^{NG}(\mathbf{y}, \mathbf{x}_\tau, \tau) \tag{14}$$

where $\tilde{K}_\tau$ is an approximation of the gain matrix K_τ introduced in (11) according to a spherical covariance prior for $\Sigma_{1|\mathbf{y}}$, we consider $\tilde{K}_\tau = \beta_\tau^2(\beta_\tau^2 + \alpha_\tau^2 \nu^2)^{-1}\mathbb{1}$ with $\Sigma_{1|\mathbf{y}} \approx \nu^2 \mathbb{1}$. Consequently, in this parameterization, neural operator $\Psi_{\tilde{\theta}}^{NG}$ shall account both for the non-Gaussian residual in (11) as well as the correction of this covariance prior. The associated training strategy involves a two-stage approach: the first stage solves (13) to train operator $\bar{\Psi}_{\bar{\theta}}$ and the second one addresses the overall training problem (12) restricted to the calibration of the residual term $\Psi^{NG}_{\tilde{\theta}}$ according to parameters $\tilde{\theta}$.

Unrolled architectures for neural operators $\bar{\Psi}_{\bar{\theta}}$ and $\Psi^{NG}_{\tilde{\theta}}$: Following neural DA frameworks (Fablet et al., 2021), we consider unrolled neural architectures based on the following generic iterative update rule for both :

$$\mathbf{x}_1^{(k+1)} = \mathbf{x}_1^{(k)} + \mathcal{R}\left(\mathbf{x}_1^{(k)}, \mathbf{y}, \mathbf{x}_\tau, \tau, k\right) \tag{15}$$

We denote by $\bar{\mathcal{R}}$ and $\mathcal{R}^{NG}$ the residual updates, respectively, for operators $\bar{\Psi}_{\bar{\theta}}$ and $\Psi^{NG}_{\tilde{\theta}}$. Whereas we use a Gaussian random initialization for $\bar{\Psi}_{\bar{\theta}}$, we consider $\left(\mathbb{1} - \beta_\tau \tilde{K}_\tau\right) \bar{\Psi}_{\bar{\theta}} + \tilde{K}_\tau \mathbf{x}_\tau$ for Ψ_θ based on (14). Regarding the design of the neural updates $\bar{\mathcal{R}}$ and $\mathcal{R}^{NG}$, we can distinguish different parameterizations of their inputs:

- ***Gradient-explicit parameterization:*** following (Fablet et al., 2021), such parameterizations generalizes gradient-based 4DVar solver (**??**) and use gradient $\nabla_{\mathbf{x}} \mathcal{J}_\tau$ as inputs for $\mathcal{R}$;
- ***Time-embedding parameterization:*** following state-of-the-art flow matching and diffusion model architectures (Lipman et al., 2023; Dhariwal & Nichol, 2021), we can consider sinusiodal embeddings for algorithm time $\tau_k = \tau + k(1-\tau)/K$. These embeddings are critical features to sample realistic and diverse samples using flow matching schemes (Dhariwal & Nichol, 2021);
- ***State-based parameterization:*** following state-of-the-art flow matching and diffusion model architectures, we can consider state $\mathbf{x}^{(k)}$ and conditionning variables $\mathbf{y}$ and $\mathbf{x}_\tau$ as inputs for $\mathcal{R}$. Such parameterizations do not exploit state-space formulation (5);
- ***Energy-informed parameterization:*** the above gradient-explicit parameterization is memory-demanding, especially at training time as it requires to differentiate two times through the operators involved in variational cost $\mathcal{J}_\tau$, first w.r.t. $\mathbf{x}$ at each iteration of the unrolling and second w.r.t. model parameters at each gradient-based update step of training scheme. To bypass this computational bottleneck, we exploit analytical gradients and subgradient approximation for energy (6). This leads to use as inputs for for $\mathcal{R}$ the following three terms: $\mathbf{y} - \mathcal{H}\left(\mathbf{x}^{(k)}\right)$, $\mathbf{x}^{(k)} - \Phi\left(\mathbf{x}^{(k)}\right)$, $(\alpha_\tau^2 \sigma_0^2)^{-1}(\beta_\tau \mathbf{x}^{(k)} - \beta_\tau^2 \mathbf{x}_\tau)$. This parameterisation explicitly benefits from the underlying variational representation. Besides, it offers a greater flexibility to the trainable unrolled solver to adapt its opitmization path and the relative contribution of these different terms along the different steps of the unrolling.
- ***Parameterization on Gaussian-vs-non-Gaussian decomposition (11):*** Following (11), we can can consider the following parameterisation for both the intialisation and the iterative residual update:

$$\left\{ \begin{array}{lcl} \mathbf{x}^{(0)} & = & \left(\mathbb{1} - \beta_\tau \tilde{K}_\tau\right) \psi_\theta(\mathbf{x}_\tau, \mathbf{y}, \tau = 0) + \tilde{K}_\tau \mathbf{x}_\tau \\ \mathbf{x}^{(k+1)} & = & \mathbf{x}^{(k)} + \dfrac{\alpha_\tau^2}{\beta_\tau} \mathcal{R}\left(\mathbf{x}^{(k)}, \mathbf{y}, \mathbf{x}_\tau, \tau\right) \end{array} \right. \tag{16}$$

  where $\tilde{K}_\tau$ is an approximation of the gain matrix K_τ introduced in (11). Assuming a spherical covariance prior for $\Sigma_{1|\mathbf{y}}$, we consider $\tilde{K}_\tau = \beta_\tau^2(\beta_\tau^2 + \alpha_\tau^2 \nu^2)^{-1} \mathbb{1}$ with $\Sigma_{1|\mathbf{y}} \approx \nu^2 \mathbb{1}$.

Besides the considered parameterization for its inputs, the selected architecture for operator $\mathcal{R}$ is another critical aspect. From the state-of-the-art in meta-learning and flow-matching (Andrychowicz et al., 2016; Hospedales et al., 2020; Dhariwal & Nichol, 2021), convolution LSTM and Unet architectures appear as relevant choices. Overall, given a selected parameterizion, the resulting sampling algorithm for $p(\mathbf{x}_1 \| \mathbf{y})$, denoted as 4DVarNet-FM, is given by Algorithm 1. Contrary to classic flow matching schemes, the unrolled architecture requires an inner loop with K iterations for each integration time step. In other words, when considering $K = 1$, the proposed scheme is similar to a classic flow matching scheme. We can also stress that applying a single integration time step, *i.e.* $N = 1$, is similar to a end-to-end inversion scheme to infer $\mathbb{E}\left(\mathbf{x}_1|\mathbf{x}_\tau, \mathbf{y}\right)$. This means that if we only need to estimate $\mathbb{E}\left(\mathbf{x}_1|\mathbf{x}_\tau, \mathbf{y}\right)$, we can run Algorithm 1 with $N = 1$ and that using $N > 1$ only allows us to sample posterior $p(\mathbf{x}_1|\mathbf{y})$ to represent the spread of the distribution around $\mathbb{E}\left(\mathbf{x}_1|\mathbf{x}_\tau, \mathbf{y}\right)$.

Algorithm 1 4DVarNet-FM algorithm to sample posterior $p(\mathbf{x}_1|\mathbf{y})$ using a Euler integration scheme

$\mathbf{x} \leftarrow GaussSampler()$
$N \leftarrow n$
for $n \leftarrow 1$ to N **do**
  $\tau \leftarrow (n-1)/N$
  $\bar{\mathbf{x}} \leftarrow \mathbf{x}$
  for $k \leftarrow 1$ to K **do**
    $\tau_k \leftarrow \tau + \frac{(k-1)}{K}(1-\tau)$
    $\bar{\mathbf{x}} \leftarrow \bar{\mathbf{x}} + \mathcal{R}\left(\mathbf{x}^{(k)}, \mathbf{y}, \mathbf{x}_\tau, \tau_k\right)$ ▷ according to the selected parameterization for $\mathcal{R}$
  end for
  $\mathbf{x} \leftarrow \mathbf{x} + \frac{1}{N}\left[\dot{\alpha_\tau}\mathbf{x} + \left(\dot{\beta_\tau} - \beta_\tau\frac{\dot{\alpha_t}}{\alpha_\tau}\right)\bar{\mathbf{x}}\right]$
end for

Contrary to DA-based solvers, we can point out that these unrolled neural architectures also apply when the parameterization of prior operator Φ is not fully known through a differential equation (Fablet et al., 2021). In such situations, we can consider a generic neural scheme for Φ. As the prior on state $\mathbf{x}_1$ in (5) relies on energy $\mathbf{x}_1 - \Phi(\mathbf{x}_1)$, auto-encoder architectures () naturally applies to Φ and Unets appear as a natural choice to deal with space-time dynamics. Other choices such as graph neural networks may also be relevant as illustrated in short-term weather forecasting (Lam et al., 2023).

Regarding the training stage, the baseline configuration is given by (**??**) such that the unrolled architecture Ψ_θ, which runs a predefined number K of iterations of update (15), approximates the conditional expectation $\mathbb{E}\left(\mathbf{x}_1|\mathbf{x}_t, \mathbf{y}\right)$. In practice, considering only an MSE (Mean Square Error) loss for state $\mathbf{x}_1$ may not fully constrain the reconstruction performance at all space-time scales. Additional MSE losses for space-time derivatives of $\mathbf{x}_1$ are often considered in this respect (Fablet et al., 2021). When prior operator Φ in (5) is not fully constrained from physics, it has to be learned from data. Following (Fablet et al., 2021), we consider the parameters of operator Φ to be a subset of parameters of the overal end-to-end operator Ψ_θ. Hence, we can complement the training loss with additional auto-encoding terms for both the reference states and the reconstructed states.

4 Evaluation framework

For evaluation purpose, we exploit and adapt the evaluation framework for model-based data assimilation introduced in () with a focus on Lorenz-63 and 2D Quasi-Geostrophic (QG) dynamical systems. These systems are representative of chaotic earth system processes and deliver up to intermediate-complexity testbeds to assess the reconstruction performance of sparsely-observed dynamical systems. We describe. below each system as well as the associated observation configuration. We also detail the considered evaluation metrics and the benchmarked schemes.

4.1 Lorenz-96 case-study

4.2 2D Quasi-Geostrophic case-study

4.3 Evaluation metrics

4.4 Benchmarked schemes

In the reported experiments, we benchmark the proposed approach with state-of-the-art model-based DA and flow matching baselines, namely:

- **4DVar (Carrassi et al., 2018)**: We consider a weak-constrained 4DVar formualtion with a fixed-step gradient descent with $xxxx$ iterations. We refer to this model-based DA scheme as 4DVar;
- **EnKF (Carrassi et al., 2018)**: We consider an ensemble Kalman Filter (EnKF) which approximates the targeted smoothing posterior through an ensemble representation of the filtering posterior using $XXXX$ members;
- **EnKS (Carrassi et al., 2018)**:
- **LEKF (Carrassi et al., 2018)**:
- **Vanilla conditional flow matching (Lipman et al., 2023)**: the classic conditional flow matching amounts to parameterizing the proposed unrolled architecture defined by (15) with a single iteration ($K = 1$) and a state-based parameterization, which does not benefit from the underlying state-space formulation. We use the same Unet backbone adapted from () as for our approach (see below). We refer to this baseline as CFM.

We refer to our approach as 4DVarNet-CFM as it bridges the end-to-end neural DA scheme introduced in (Fablet et al., 2021) and conditional flow matching (Albergo et al., 2025; Lipman et al., 2023). We distinguish the benchmarked 4DVarNet-CFM configurations according to:

- The number K of unrolling steps, denoted as $\times K$;
- The exploitation of the unrolling scheme (16) informed by the decomposition (11) according to Gaussian and non-Gaussian components. We use suffix $+GnG$ to configurations leveraging this parameterization;
- The use of time-embedding T, observation-explicit O, state-explicit S, gradient-explicit (∇) and/or ernegy-based input parameterizations for the the residual update operator $\mathcal{R}$. For instance, suffix $+TS\nabla$ refers to combining time-embedding, state and gradient inputs for $\mathcal{R}$ in (15), *i.e.* $\mathcal{R}\left(\bar{\mathbf{x}}_\tau^{(k)}, \mathbf{y}, \mathbf{x}_\tau, \tau\right) = \mathcal{R}\left(\bar{\mathbf{x}}_\tau^{(k)}, \nabla_{\mathbf{x}_1}\mathcal{J}_\tau(\mathbf{x}_1, \mathbf{y}, \mathbf{x}_\tau), \mathbf{x}_\tau, \tau\right)$.

Overall, $en4DVarNet-CFMx10+GnG+TOS$ would for instance refer to the proposed approach with a 10-iteration unrolling based on (16) using time-embedding, observations and state as inputs to residual update operator $\mathcal{R}$.

5 Results

5.1 Benchmarking the estimation of the conditional mean $\mathbb{E}\left(\mathbf{x_1}|\mathbf{y}\right)$

5.2 Benchmarking the estimation of the conditional mean $\mathbb{E}\left(\mathbf{x_1}|\mathbf{y}\right)$

5.3 Robustness to modelling uncertainties

6 Discussion

7 Impact statement

The proposed 4DVarNet-FM framework introduces a principled, physics-informed approach to neural data assimilation that bridges variational formulation and conditional flow matching.

Scientific Advancement: By embedding physical priors directly into the architecture via trainable stochastic interpolants, our method offers a robust path to sampling posteriors in high-dimensional, non-linear Earth system dynamics where traditional methods (e.g., EnKF) may struggle with non-Gaussianity or computational tractability. Methodological Versatility: This work moves beyond toy examples, providing a general-purpose neural solver that can be adapted to various spatio-temporal reconstruction tasks, potentially reducing the development cycle for operational data assimilation systems.

Sustainability & Reliability: By leveraging energy-informed parameterization, the framework achieves computational efficiency without sacrificing physical consistency, facilitating the adoption of machine learning in mission-critical applications such as ocean monitoring and global weather forecasting.

Ethical Considerations: We recognize that automated data assimilation systems in geosciences impact societal preparedness for environmental shifts. Our framework improves the reliability of these systems, but as with all AI-native methods, future deployments should involve rigorous validation against independent observations to ensure the structural integrity of the generated forecasts.

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